Uhlenbeck Integrals, a Rational Lax Matrix, and Compact Liouville Fibers for the Neumann Sphere

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Abstract

For the anisotropic Neumann oscillator on the round two-sphere we prove global completeness and derive all three Uhlenbeck integrals with their two linear relations.

1 Frozen constrained system and theorem

Let $A = \mathrm{diag}(a_1, a_2, a_3)$ with $a_1 < a_2 < a_3$. On

$$T^*S^2 = \{(x, p) \in \mathbb{R}^3 \times \mathbb{R}^3 : |x|^2 = 1,\ x \cdot p = 0\}$$

take

$$H = \frac{1}{2}(|p|^2 + x^T Ax), \qquad \dot{x} = p, \qquad \dot{p} = -Ax + \alpha x, \quad \alpha = x^T Ax - |p|^2. \tag{1}$$

Put $L_{ij} = x_i p_j - x_j p_i$ and

$$F_i = x_i^2 + \sum_{j \neq i} \frac{L_{ij}^2}{a_i - a_j}. \tag{2}$$

Theorem 1. *The flow* (1) *is complete. The F_i are conserved, pairwise Dirac–Poisson commuting, and satisfy*

$$\sum_i F_i = 1, \qquad \sum_i a_i F_i = 2H.$$

2 Completeness and the Uhlenbeck identities

The two constraint derivatives are

$$\frac{d}{dt}|x|^2 = 2x \cdot p, \qquad \frac{d}{dt}(x \cdot p) = |p|^2 - x^T Ax + \alpha = 0$$

on T^*S^2. Also $\dot{H} = 0$. On $H = E$ the identity $|p|^2 = 2E - x^T Ax$ bounds p; the orbit remains in a compact subset, and the smooth vector field is complete.

The angular momenta satisfy

$$\dot{L}_{ij} = (a_i - a_j) x_i x_j. \tag{3}$$

Hence

$$\dot{F}_i = 2x_i \left(p_i + \sum_{j \neq i} L_{ij} x_j \right) = 0,$$

because the parenthesis equals $p_i + x_i(x \cdot p) - p_i |x|^2$. Pairing (i, j) and (j, i) proves $\sum F_i = 1$. The weighted pairing and the Gram identity give

$$\sum_i a_i F_i = x^T Ax + \sum_{i<j} L_{ij}^2 = x^T Ax + |p|^2 = 2H.$$

This is the *compact Uhlenbeck owner* added in revision round zero.

Source boundary

Neumann owns the original mechanical problem; Moser owns the modern quadrics/spectral lineage; Knoerrer owns the geodesic correspondence cited here. This convention-locked reconstruction makes no priority claim.

References

[1] C. Neumann, "De problemate quodam mechanico, quod ad primam integralium ultraellipticorum classem revocatur," *J. Reine Angew. Math.* 56 (1859), 46–63, DOI `10.1515/crll.1859.56.46`.

[2] J. Moser, "Geometry of Quadrics and Spectral Theory," in *The Chern Symposium 1979*, Springer (1980), 147–188, DOI `10.1007/978-1-4613-8109-9_7`.

[3] H. Knoerrer, "Geodesics on quadrics and a mechanical problem of C. Neumann," *J. Reine Angew. Math.* 334 (1982), 69–78, DOI `10.1515/crll.1982.334.69`.